北师香港浸会大学
BEIJING NORMAL-HONG KONG BAPTIST UNIVERSITY

MATH1003 Linear Algebra

Instructor: Dr. Xinwei Du
Faculty of Science and Technology
Beijing Normal-Hong Kong Baptist University



Instructor and TA information



- Instructor: Dr. Xinwei Du
  - xinweidu@bnbu.edu.cn
  - T6-302-R9

0703

- TA: Ms. Ziyuan Liu and Mr. Qingshan Wang
- T29-402-12, liuziyuan@bnbu.edu.cn

Fri. 5-6 pm.

Timetable



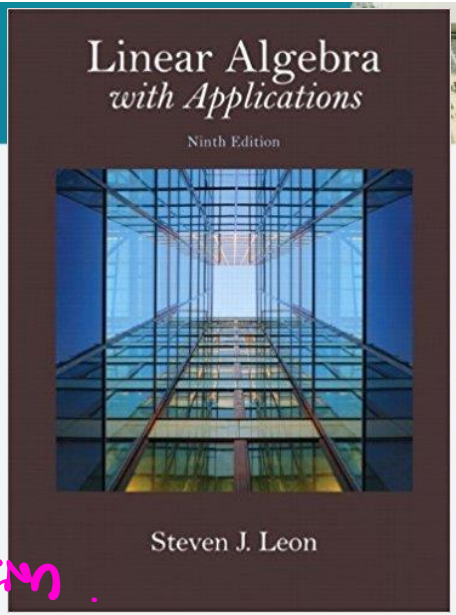
Timetable for 2026-2027 Semester 1

T6-302-R9

Time	Monday	Tuesday	Wednesday	Thursday	Friday
09:00 - 09:50		Office Hour *UG Meeting*		Office Hour	Office Hour
10:00 - 10:50		Office Hour *UG Meeting*		Linear Algebra (CDS) T5-202	Office Hour
11:00 - 11:50				Linear Algebra (CDS) T5-202	
12:00 - 12:50			*LUNCH BREAK*		
13:00 - 13:50		Office Hour		Office Hour	Linear Algebra (CDS) T7-503
14:00 - 14:50		Linear Algebra I (STAT) T29-602			Office Hour
15:00 - 15:50		Linear Algebra I (STAT) T29-602			Linear Algebra I (STAT) T4-603
16:00 - 16:50		Office Hour			Office Hour

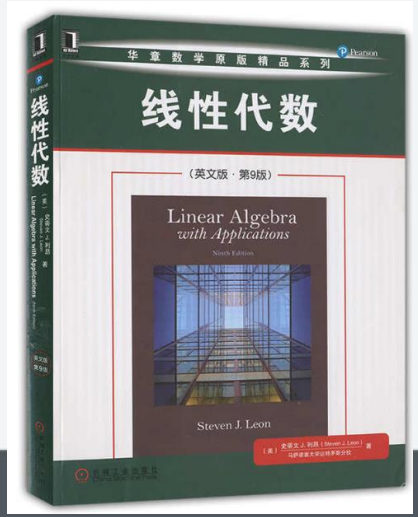
Textbook

- Linear Algebra with Applications
  - Eighth/Nineth Edition by **Steven J. Leon**



8.9. 10th edition.

- Reference video:
  - MIT open course by Prof. Gilbert Strang
  - http://ocw.mit.edu/courses/mathematics/18-06-linear-algebra-spring-2010/video-lectures/
  - Khan academy by Sal Khan: https://www.khanacademy.org/math/linear-algebra
  - 3Blue1Brown, Essence of Linear Algebra: https://www.bilibili.com/video/BV1ys411472E
- http://immersivemath.com/ila/index.html



Assessment

13. ASSESSMENT METHODS (AMS)

Type of Assessment Methods	Weighting	CILOs to be addressed	Description of Assessment Tasks
Assignments	20%	1-4	Assignments will test the students' understanding on theories and ability to use theories to solve problems.
Class exercises	20%	1-2	Exercises will test the students' understanding on theories which are just taught by teachers.
Projects	20%	1-4	Projects will test students ability to use some basic tools to solve some simple but practical problems.
Final examination	40%	1-4	The final examination will check how well the students understand the basic theories in this course and apply the theories to the real problems.

continuous assessment.

1. Assignments 15% & Attendance 5%

1> 1 assignment per week (4-6 questions)

2> Online submission via iSpace.

1 pdf file.

3> late submissions

2. Class exercises 20%

↙ ↘

Quiz 1 10% → Week 6 10.12 - 10.16

Quiz 2. 10% → Week 13 11.30 - 12.4

50 min.

Notify 1 week in advance.

3. Projects 20%

↓

Mid-term Exam. 2hrs.

2026.11.2 - 2026.11.8. 7-9 pm. ← Upto Ch3.2

11.5 7-9pm (3 persons) 11.7 2-4pm (1 person)

11.6 7-9 am (3 persons) 11.8 9-11 am 2-4pm

4. Final exam 4% 2.5 hrs.

{ Final exam ≤ 20'
or
weighted score ≤ 30' } > fail

Assignments



- Individual assignments weekly
  - Written questions
  - Do not collaborate with other students
  - The deadline is non- negotiable
  - Late submission will get a **penalty**

Late Submission	Mark Reduction
$\leq$ 12 hrs	20%
$\leq$ 24 hrs	40%
$>$ 1 day	100%

- Academic honesty
  - Copying assignments will get a **ZERO** grade

What is linear algebra?



- **Linear algebra** is a branch of mathematics concerned with the study of vectors, with families of vectors called *vector spaces* or *linear spaces*, and with functions that input one vector and output another, according to certain rules.

ch4.

ch3 ☆☆☆

- **Linear algebra** is central to modern mathematics and its applications.

Linear Algebra

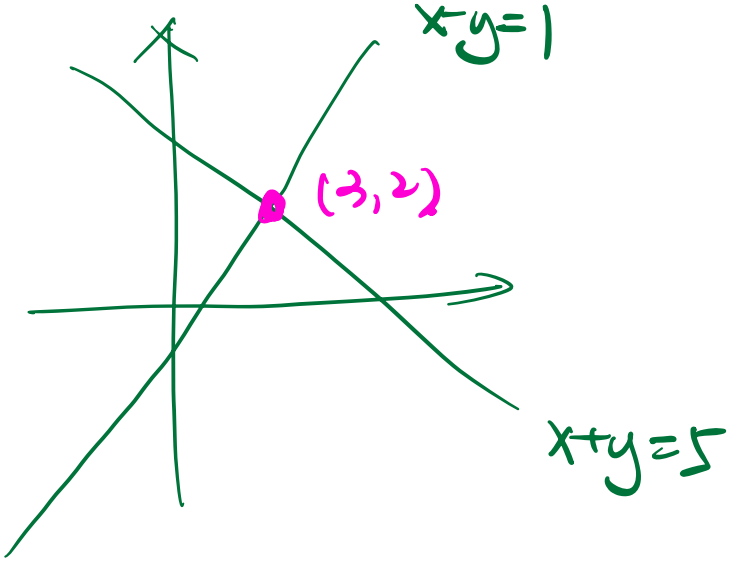
Linear relationship / operations.

- Scalar multiplication
- Addition.

$3x + 2y = 5$

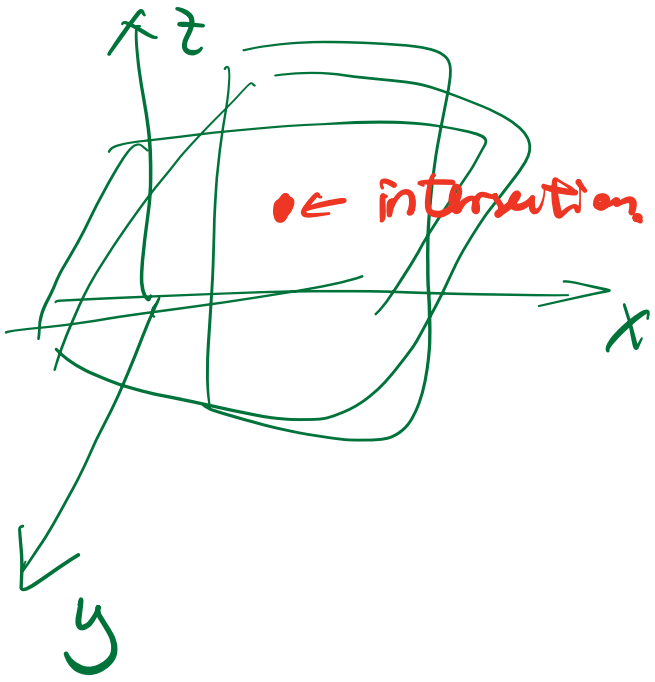
2-D

Eg. ① $\begin{cases} x+y=5 \\ x-y=1 \end{cases} \Rightarrow \begin{cases} x=3 \\ y=2 \end{cases}$



3-D

② $\begin{cases} x+y+z=6 \\ 2x+y+z=18 \\ 3x-2y+5z=20 \end{cases}$



Extend to a general case.

n variables: $x_1, x_2, \cdots, x_n$

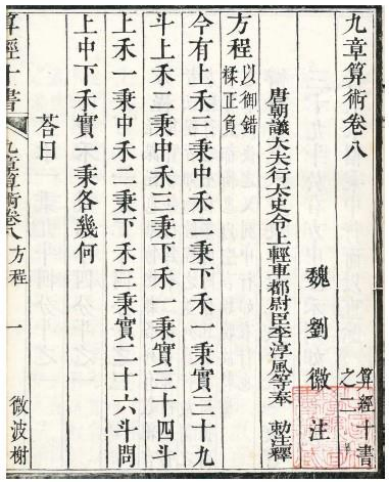
m equations.

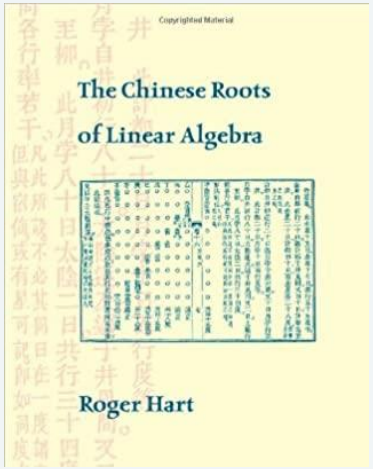
How many solutions? All.

Example of systems of equations

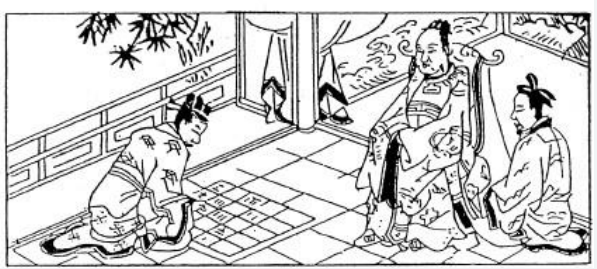


Liu Hui
3rd Century AD





Problem 1, Chapter 8 of Jiu Zhang Suan Shu



$$\begin{cases} 3x+2y+z=39 \\ 2x+3y+z=34 \\ x+2y+3z=26 \end{cases}$$

Example of systems of equations



$$\begin{cases} 3x+2y+z=39 \\ 2x+3y+z=34 \\ x+2y+3z=26 \end{cases}$$

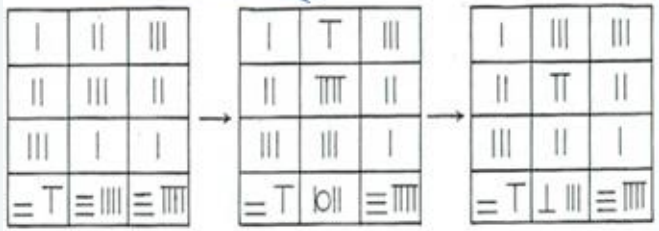
1,2,3,4,5,6,7,8,9

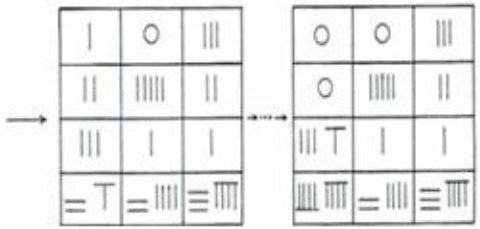


1	2	3
2	3	2
3	1	1
26	34	39

0	0	3
4	5	2
8	1	1
39	24	39

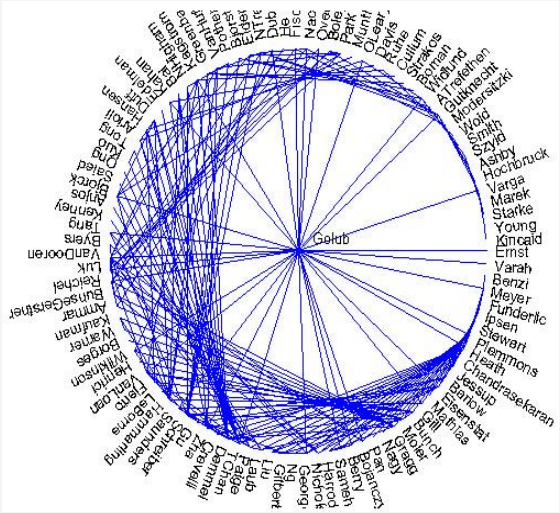
0	0	3
0	5	2
36	1	1
99	24	39

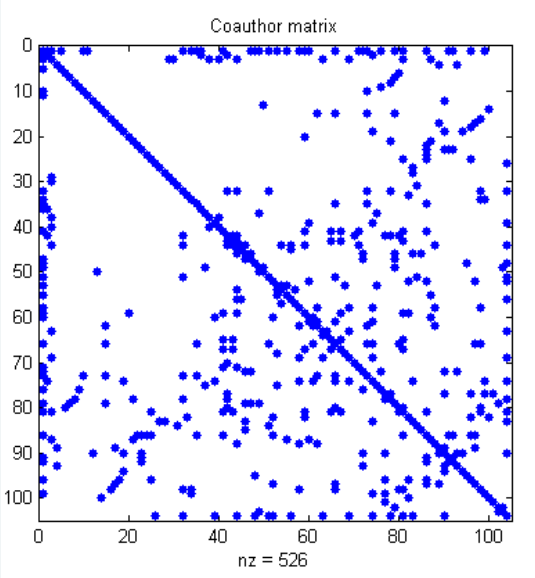




Network analysis

- Social network
- Economy network
- WWW
- Protein rank
- Gene rank
- Electrical network
- Coauthor network



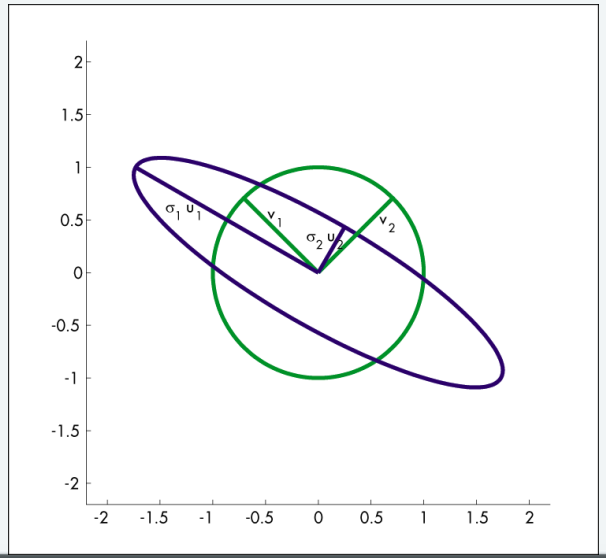


Singular Value Decomposition

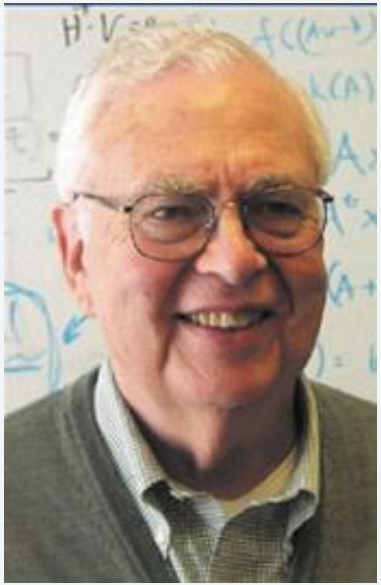
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Singular Value Decomposition, Eigenfaces, and 3D Reconstructions*









Linear Algebra is the foundation of data science and AI!

How to learn Linear Algebra?



- **Preview before attending each class;**
- **Have a pencil and scratch paper ready. Practice on what you are reading. *Mathematics is not read like a novel* ;**
- **Make sure you understand the methods for solving the examples done in class; "*If you can't explain it, you don't understand it*".**
- **Identify ideas that were discussed in the lectures and work out the additional examples presented in the book.**
- **Finish your homework carefully on time.**

Study Tips



- Attend **ALL** the classes!
- Read (and sometimes write) the lecture notes.
- Spend at least 3 out-of-class hours each week.
- Use the discussion group on iSpace, Wechat.
- Complete the assignments individually.
- All in all, work hard!
- And have fun!